

NAGA DATHA SAIKIRAN BATTULA

PhD, Data Science

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📍 Harrison, New Jersey

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SUMMARY

Aspiring researcher with a strong background in Natural Language Processing, architectures of LLMs, and data science fundamentals. Proven ability to develop and deploy innovative, data-driven solutions that enhance the use of conversational AI in real-time, dynamic systems.

EXPERIENCE

Doctoral Researcher

NJIT's Intelligible Information Visualization (NiiV) lab

📅 01/2025 – Present

📍 Newark, NJ

- Developed a interactive Human-AI conversational interface for real-time safeguarding of power grids.
- Implemented attribute extraction pipeline for multimodal data (text and images) analysis and question-answering in science domains.

Graduate Research Assistant

NJIT's Intelligible Information Visualization (NiiV) lab

📅 02/2024 – 12/2024

📍 Newark, NJ

- Worked on validation of model-based simulations of Weather Research & Forecast (WRF) Model by steering with different Climate Science based physics parameters.

PUBLICATIONS

📄 Submitted

- N. D. S. Battula, C. Fitzpatrick, S. A. R. Naqvi, T. Yin, S. P. nandanoori, K. Bhattacharjee, S. Kundu, and A. Dasgupta, "Divide and concur: Design study of a multimodal, real-time human-ai collaboration framework for safeguarding powergrids," EuroVis, 2026.

👥 Conference Proceedings

- N. D. S. Battula, H. R. Kambhampaty, Y. Vijayalata, and R. N. Ashlin Deepa, "Deep-learning residual network based image analysis for an efficient two-stage recognition of neurological disorders," in *2023 2nd International Conference for Innovation in Technology (INOCON)*, 2023, pp. 1–6. DOI: 10.1109/INOCON57975.2023.10101037.
- H. R. Kambhampaty, N. D. S. Battula, S. Rentala, S. C. Challa, and A. K, "Network optimization of large-scale cnns for applications in autonomous machines," in *2023 2nd International Conference for Innovation in Technology (INOCON)*, 2023, pp. 1–6. DOI: 10.1109/INOCON57975.2023.10101330.
- H. R. Kambhampaty, N. D. S. Battula, Y. Vijayalata, and A. D. R. N, "A novel deep-learning based classification of alzheimer's disease in adults," in *2022 IEEE Delhi Section Conference (DELCON)*, 2022, pp. 1–6. DOI: 10.1109/DELCON54057.2022.9752991.

EDUCATION

Ph.D. in Data Science

New Jersey Institute of Technology

📅 Jan 2025 – present

M.S. in Data Science

New Jersey Institute of Technology

📅 Sep 2023 – Dec 2024

GPA: 3.91/4.0

MS Thesis: Surveying the role of visual analytics in human-machine teaming
<https://digitalcommons.njit.edu/theses/2946/>

B.Tech. in Computer Science and Engineering

Gokaraju Rangaraju Institute of Engineering and Technology

📅 Aug 2019 – May 2023

GPA: 9.53/10

SKILLS

Generative AI

- Architectures & Techniques: Retrieval Augmented Generation (RAG), Agentic AI, LLM Fine-Tuning, Prompt Engineering, Chain-of-Thought
- Frameworks & Libraries: HuggingFace Transformers, OpenAI API, LangChain

Machine Learning

- Machine Learning Techniques: Reinforcement Learning, Dynamic Programming, Natural Language Processing (NLP)
- Tools: Tensorflow, Pytorch

Programming Languages and Tools

Python (libraries: numpy, pandas, TextBlob, NLTK), C, C++, Java

Databases: Postgres, MongoDB, MySQL

Data Visualization: Tableau, Matplotlib, Seaborn, D3.js

CLOUD & MLOps

- Cloud Platforms: AWS (EC2, SageMaker, S3, Route 53), Google Cloud (Compute Engine, Vertex AI)
- Infrastructure: Docker, Virtual Machines, CI/CD Pipelines